Richards Britto

Software Engineer

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ABOUT ME

Software engineer building REST APIs, and AI applications. Passionate about scalable software, LLM and generative model architectures, and the full computing stack from hardware to software.

EDUCATION

Master of Science | Uppsala University

Sept 2021 – Nov 2023 Uppsala, Sweden

- Major: Image Analysis and Machine Learning

Bachelor of Technology | NIT Trichy

Sept 2016 – May 2020 Tiruchirappalli, India

- Major: Electronics and Communication Engineering
- Minor: Humanities and Social Sciences (Economics)
- GPA: 8.01

TECHNICAL SKILLS

Language Python, C++, HTML/CSS, LaTex | **Backend** FastAPI, pydantic, httpx, ffmpeg | **DevOps** Azure, Vertex-AI, Docker, CI/CD, Git, Linux, bash | **AI** PyTorch, LangGraph, diffusers, transformers, OpenCV, Open3D, NumPy, Pandas, Ollama, Matplotlib | **Devlopment** Agile, VSCode, Jira, Confluence, Github Copilot, Cursor, Opencode, MATLAB, Blender

WORK EXPERIENCE

AI & 3D Software Developer | Inter IKEA Group

August 2024 – CURRENT Älmhult, Sweden

- Developed **REST APIs** for a **FastAPI**-based backend service for IKEA's **AgentVerse** platform with plugin discovery, secure file handling, **Magnific AI** image/video generation (10 models), and **docling-serve** based file parsing.
- Implemented **CI/CD** deployment pipelines on **Azure**, integrated JWT authentication, centralized logging with **Splunk**, and improved deployment reliability.
- Evaluated SOTA generative AI models across image, video, and 3D (**Qwen-VL, WAN, LTX, Hunyuan3D, Trellis, Kling, Seedance, Gemini**, etc) for production use cases.
- Worked with cross-functional **Agile teams** to deliver production software, including a **Vertex-AI** based agent that generated ~60K images, increasing sales and reducing 3D production costs.

Computer Vision Engineer Intern | Vidhance AB

Sept 2022 – Jan 2023 Uppsala, Sweden

- Built **Blender Python API** tool simulating real-world camera effects and sensor motion, saving 20+ hrs/week; collaborated in **agile team** to generate videos with depth map, flow map, and customizable environments.

ACADEMIC RESEARCH

iNPH Diagnosis using DL (Masters Thesis) |

Feb 2023 – Sept 2023 Uppsala, Sweden

- Estimated 3D biomarker voxels with sub-mm accuracy via **nn-UNet**, aligned MRIs to Talairach space, and computed Evan's Index using FastSurfer on 3090-Ti (Linux).

Occlusion Aware Origami Pose Estimation |

May 2020 – April 2021 NUS, Singapore

- Achieved 97% pose estimation accuracy using a custom **Mask-RCNN+cGAN** pipeline combining RGB-D for depth and 3D keypoint generation.

PROJECT WORK

3D Object Detection using LiDAR |

Feb 2024 – Mar 2024 Uppsala, Sweden

- Implemented SFA3D algorithm for 3D object detection on the **KITTI dataset**, utilizing Bird's Eye View representation of LiDAR point cloud data. Employed **Keypoint-FPN** architecture with **ResNet 18** backbone.

MIP Control and Dynamcs |

Dec 2023 – Feb 2024 Uppsala, Sweden

- Designed a control system for Mobile Inverted Pendulum in **MATLAB**, incorporating **Extended Kalman Filters** for state estimation and **PD controllers** for dynamic stabilization, resulting in 25% increase in system stability.

NeRF and Gaussian Splatting |

Dec 2023 – Jan 2024 Uppsala, Sweden

- Employed Blender's Python API to generate a synthetic dataset containing ∼150 RGB images, camera parameters, along with meta data, then trained **NeRF** and **Gaussian Splatting** to model volumetric scenes.

Monocular ORB-SLAM (MATLAB) |

Nov 2023 – Dec 2023 Uppsala, Sweden

- Executed **6 DoF** camera pose estimation and 3D scene reconstruction utilizing the TUM RGBD dataset.
- Leveraged the **PnP** algorithm for pose estimation, refined results through **bundle adjustment**, and applied **Bags-of-Words** for loop closure, 0.12 RMSE was achieved.

Ray Tracing and Rasterization |

Sept 2023 – Oct 2023 Uppsala, Sweden

- Engineered rasterization, ray tracing techniques in C++ including motion blur, textures, and lights, achieved a 40% rendering speed boost using BVH.